

MA 002 Preparatory Mathematics - 2

Course Code: MA 002

Course Name: Preparatory Mathematics - 2

L-T-P-C: 3-1-0-4

Students Intended for: Preparatory Students

Core or Elective: Core

Approval: 50th BoA

Course Contents:

- **Functions and Their Graphs:** Functions, Inverse Function, Elementary Function and their Graphs. [6 Lectures]
- **Differentiation:** Limit, Continuity. Derivative and its geometrical significance. Derivatives of sum, difference, product and quotient of functions. Derivatives of polynomial, rational, trigonometric logarithmic and exponential functions. Differentiation of composite and implicit functions. [12 Lectures]
- **Applications of Derivatives:** Tangents and normal. Increasing and decreasing functions. Maxima and Minima. [6 Lectures]
- **Integration and its Applications:** Integration as the inverse process of differentiation. Integration by parts and by substitution. Definite integral and its application to determination of areas (simple areas). [8 Lectures]
- **Differential Equations:** First order ordinary differential equations. Homogeneous and exact equations. First order linear equations.

Text Books:

1. G. B. Thomas and R. L. Finney, **Calculus and Analytical Geometry**, Addison Wesley / Narosa.
2. T. M. Apostol, **Calculus**, 2nd Edition, Wiley Eastern.

Reference Books:

1. R. K. Jain and S. R. K. Iyengar, **Advanced Engineering Mathematics**, 3rd Edition, Narosa Publisher.